monitoring at the Comprehensive Epilepsy Center at New York University Langone Health Center, Brigham and Women's Hospital, Children's Hospital Boston, and Johns Hopkins Medical School and University of Wisconsin School of Medicine and Public Health.

Electrophysiological data acquisition. Brain activity will be recorded from intracranially implanted subdural platinum-iridium electrodes embedded in SILASTIC sheets (2.3 mm diameter contacts, Ad-Tech Medical Instrument and PMT Corporation). The decision to implant, electrode targeting, and the duration of invasive monitoring will be solely determined on clinical grounds and without reference to this or any other study. Macroelectrodes will be arranged as grid arrays (8 × 8 contacts, 10 or 5 mm center-to-center spacing), linear strips (1 × 8/12 contacts), or depth electrodes (1 × 8/12 contacts), or a combination thereof. Subdural electrodes will cover extensive portions of lateral and medial frontal, parietal, occipital, and temporal cortex of the left and/or right hemisphere. Recordings from grid, strip and depth electrode arrays will be made using a Natus Quantum (Pleasonton, CA), Xltek (San Carlos, CA) or a Blackrock system (Salt Lake City, UT) amplifier. Recordings are obtained continuously during the patients' stay in the hospital. All data will be stored with stimulus and timing markers permitting offline synchronization.

Surface reconstruction and electrode localization. Pre-surgical T1-weighted MRIs (no electrodes) and post-surgical CT scan (with electrodes) will be acquired for each patient and used to determine electrode locations, following the general methodology of Yang et al. (2012) or Dale et al. (1999) [62, 63] using the Freesurfer package (HU). For NYU, skull-stripped post-surgical CT images will be linearly co-registered to the pre-surgical MRI. Electrode locations will be extracted using the co-registered CT images, and projected to the reconstructed brain surface, generally following the procedure described by Yang and colleagues. MRI images will be nonlinearly registered to an MNI-152 template. The same transformation will be applied to the co-registered CT image in order to map the extracted electrode coordinates in Montreal Neurological Institute (MNI) space. A three-dimensional reconstruction of each patient's brain will be computed using FreeSurfer (http://surfer.nmr.mgh.harvard.edu).

At Harvard, the pial surface of each subject will be reconstructed using Freesurfer from the presurgical MRI. The Freesurfer function "recon-all" will be used, following the fully-automated directive ("-autorecon-all" flag). The CT images of the implanted electrodes will be localized to the MRI reconstruction using the iELVis package (Groppe et al., 2017) [64]. CT gantry tilt will be corrected using the dcm2niix package from www.nitrc.org. Electrode grid and strip orientation will be identified in the CT scan based on pre-surgical sketches and platinum marker guides (Ad-Tech, Racine, WI, USA). The electrode locations will then be summarized by mapping them onto one of 36 brain areas based on the parcellation of the Desikan-Killiany atlas (Desikan et al., 2006) [65] using the "recon-all" function in Freesurfer.

Behavioral setup. The experiment will be controlled and the stimuli will be presented on a Dell Precision 5540 laptop, with 15.6" Ultrasharp screen (screen size 357.27 x 235.47 mm2; resolution 1920x1080) running on Windows 10. The laptop will be positioned at 55–65 cm from the patient. Audio will be presented through loudspeakers.

Eye tracker. Eye-tracking data will be collected throughout the duration of the experiment using a Tobii-4C eye-tracker (New York University) and EyeLink 1000 Plus Camera (Harvard University and University of Wisconsin). Thirteen-point calibration will be performed several times during the experiment. The calibration will be performed at the beginning of the experiment, and recalibrated if necessary to meet precision requirements at the beginning of each mini-block. Pupil size and corneal reflection data will be collected at a sampling rate of 90Hz at New York University and at a sampling rate of 500Hz at Harvard university and University of Wisconsin. Only one eye will be recorded as determined by ocular dominance. The experiment will not be influenced by the Eye-tracking recording.

Data management and sharing plan

Our plan to share neuroimaging, electrophysiological, behavioral and eye tracking data, and our management of intellectual property will be in accordance with the policies and guidelines of our institutions and NIH. All investigators involved in this project will adhere to NIH's Data Sharing Policy and Implementation Guidance of March 5, 2003 and NIH Grants Policy on Sharing of Unique Research Resources including the "Sharing of Biomedical Research Resources: Principles and Guidelines for Recipients of NIH Grants and Contracts" issued in December, 1999.

The investigators acknowledge their willingness to share data and materials stemming from this project in order to maximize impact and to accelerate discoveries to understand consciousness and its neural substrates. As soon as data have been acquired, data will be uploaded to the central cloud server of the project, hereby making it immediately available to all members of the team. Throughout the course of the 3 years that the project is expected to last, data will be shared with local colleagues at seminars and talks, and with the scientific community at large by posters and presentations at local, regional, national and international scientific meetings. Finally, data will be presented by publication to the widest audience possible.

Following publication of the primary adversarial collaboration manuscript, the research data will be made available to the scientific community through public repositories.

Specifically, as part of our resource sharing plan, we will develop a website on which the following resources pertaining to this proposal will be made publicly available:

1. Any Matlab (binaries)/ Python code used to analyze or pre-process the data
2. Any MRI pulse sequences (binaries) used to acquire the data
3. All (de-identified and PHI removed) MRI, neurophysiological, eye tracking and behavioral data adhering to the FAIR principle. Specifically, raw data will be shared both in native formal and also in Brain Imaging Data Structure BIDS format (http://bids.neuroimaging.io). All metadata will be made available.
4. Any peer-reviewed article (depending on journal restrictions)
5. Any experimental code

In order to access or download files containing behavioral, eye tracking, neurophysiological or MRI data, users will have to register. As they register, they will agree to restrictions against attempting to identify study participants, restrictions on redistribution of the data to third parties, and to properly acknowledge the data resource. The neurophysiological recordings (clinical iEEG and experimental electrodes), relevant task data, electrode coordinates in MNI space and essential, de-identified clinical data using NINDS Common Data Elements (age, sex, duration of epilepsy, epilepsy etiology, preoperative imaging findings) and schematics of seizure onset areas will also be made available. Data will be converted to shareable data formats (BIDS). Machine-readable annotation of the task and Matlab/Python scripts to synchronize data streams will be included. All data will be reviewed prior to upload to ensure they contain no PHI. Data will be stripped of voice recordings and be HIPAA compliant. Data will be made available to the research community by sharing them through different steps in the analysis pipeline. We will share un-processed (DICOM), minimally preprocessed (NIFTI format) and final processed (NIFTI format) data. All custom code and analysis pipeline will also be shared.

Discussion

We have embarked upon an ambitious and large-scale endeavor involving a dozen laboratories in three continents using an open science and adversarial collaborative process. The field has

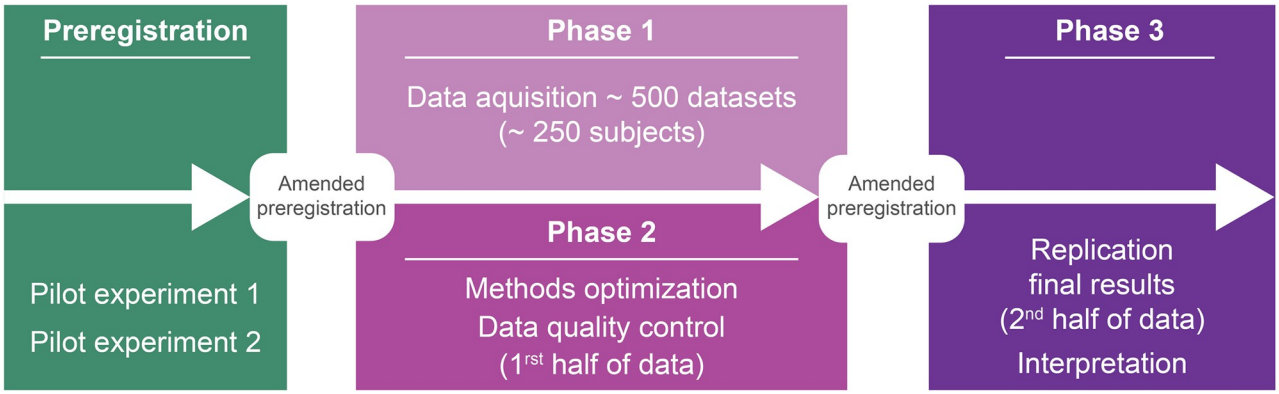

Fig 4. Timeline for the adversarial collaboration protocol. Number of subjects refers to the sum of subjects collected in the two experiments. Due to the COVID-19 pandemic of 2020–2021, phase 1 and 2 will be carried out in parallel and adjustments to the timing of some of the phases may be necessary.

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already made substantial progress with a myriad of individual studies [e.g., 18, 66], but systematic tests of the theories' predictions in a large-scale, multimodal design have not yet been attempted. The present study protocol results from the first stage of this project–finding agreement among scientists with conflicting views for two meticulously planned studies, the associated instruments and analysis tools, as well as funding for such a costly enterprise. The two studies were extensively piloted (on more than 250 subjects; see OSF page). Those pilots established the adequacy of the experiments for yielding the desired behavioral and eye movement patterns, and served to rule out potential confounds. We are now in the process of collecting 500 datasets (250 subjects x 2 studies) using our chosen techniques—fMRI, M-EEG and iEEG—in phase 1 (Fig 4). This will complete the data acquisition phase. The first half of this massive data volume, that will reside in a secure server during the completion of the project, will be used to test the specific and detailed hypotheses described above, in phase 2. All aspects of the processing pipeline will be finalized and frozen before proceeding to the analysis of the second half of the data in phase 3, to assure the generalizability of the final results. These will then be published in platform papers, with all data, meta-data, stimulation protocols, and analysis code made freely available to anyone for further analysis or extension experiments.

GNW and IIT make several different fundamental assumptions about consciousness that are, at this point, challenging to directly test at the relevant level of granularity. In the two experiments outlined here, we focus on specific predictions that differentiate between these two theories regarding various aspects of the NCC that can be empirically addressed using contemporary brain recording techniques. Given the vast and ill-understood complexity of the brain, extant instrumental and biological variability across subjects and trials, and the distinct acquisition methods used, it is possible that no unambiguous answer may emerge from these experiments. At the least, we expect our results to challenge specific aspects of either or both theories and to further our understanding of conscious visual perception by tracking its footprints in the human brain.

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